Multivariable Regression

Part 2 - inference, transformations, and other loose ends

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Introduction

We recently learned about *one-way ANOVA*, a statistical test for scenarios involving a nominal categorical explanatory variable and a quantitative outcome. The alternative model in one-way ANOVA can be expressed as a regression model:

Group	one-way ANOVA	Regression
1	$y_i = \mu_1 + \epsilon_i$	
2	$y_i = \mu_2 + \epsilon_i$	$y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i$
3	$y_i = \mu_3 + \epsilon_i$	

X_1 and X_2 are dummy variables (0's and 1's), such that:

- $\mu_1 = \beta_0$ (the reference group when both X_1 and X_2 are zero)
- $\mu_2 = \beta_0 + \beta_1$ (when $X_1 = 1$)
- $\mu_3 = \beta_0 + \beta_2$ (when $X_2 = 1$)

The F-test for Regression

Similarly, the null model in one-way ANOVA can also be expressed as an *intercept-only* regression model:

Group	one-way ANOVA	Regression
All	$y_i = \mu + \epsilon_i$	$y_i = \beta_0 + \epsilon_i$

- Thus, the F-test performed in one-way ANOVA is actually a comparison of *nested regression models*
  - Two models are nested when one model, known as the *reduced model*, is a special case of a larger model
  - $y_i = \beta_0 + \epsilon_i$ is a special case of the larger model: $y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i$ that occurs when $\beta_1 = 0$ and $\beta_2 = 0$

The F-test for Regression

In regression, the F-test's null hypothesis is that additional predictors in the larger model provide *no improvement* over the predictors already present in the reduced model, or in statistical symbols:

$$H_0 : \beta_j = \beta_{j+1} = \beta_{j+2} = \ldots = \beta_p = 0$$

- $\{\beta_j, \beta_{j+1}, \ldots, \beta_p\}$ are the respective coefficients of variables in the larger, alternative model that are *not present* in the reduced model.
  - Thus, *both models* contain $\{\beta_1, \beta_2, \ldots, \beta_{j-1}\}$ and we don't learn anything about the corresponding variables from the test

The F-test for Regression

The F-test assesses whether the *sum of squared* residuals decreases by more than would be expected by chance when additional predictors are included in a regression model:

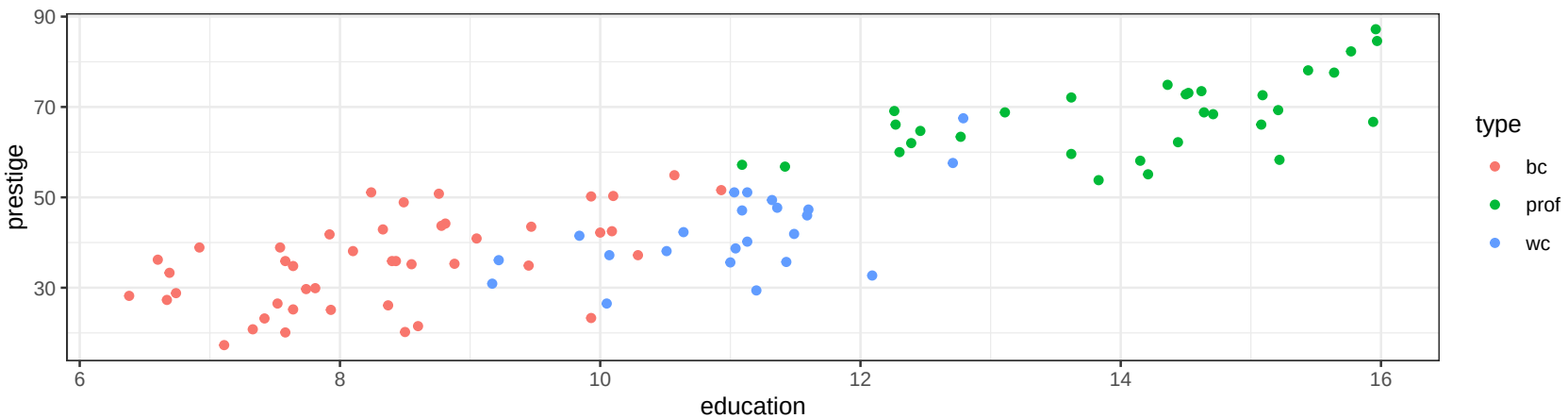
$$F = \frac{(SS_0 - SS_1)/(d_1 - d_0)}{SS_1/(n - d_1)}$$

- Here SS_0 is the sum of squares of the *reduced model*, which is the smaller one that represents the null hypothesis
- SS_1 is the sum of squares of the *larger model*, which represents the alternative hypothesis
- d_0 is the number of parameters in the reduced model, while d_1 is the number of parameters in the larger model

Example - Modeling Occupational Prestige

A study collected data on $n = 98$ occupations. We'll consider the variables:

- **prestige**: the average prestige rating of the job (from 0 to 100)
- **education**: the average number of years of schooling for people holding the job
- **type**: the type of job, either skilled professional (prof), blue collar (bc), or white collar (wc)



Example (cont.)

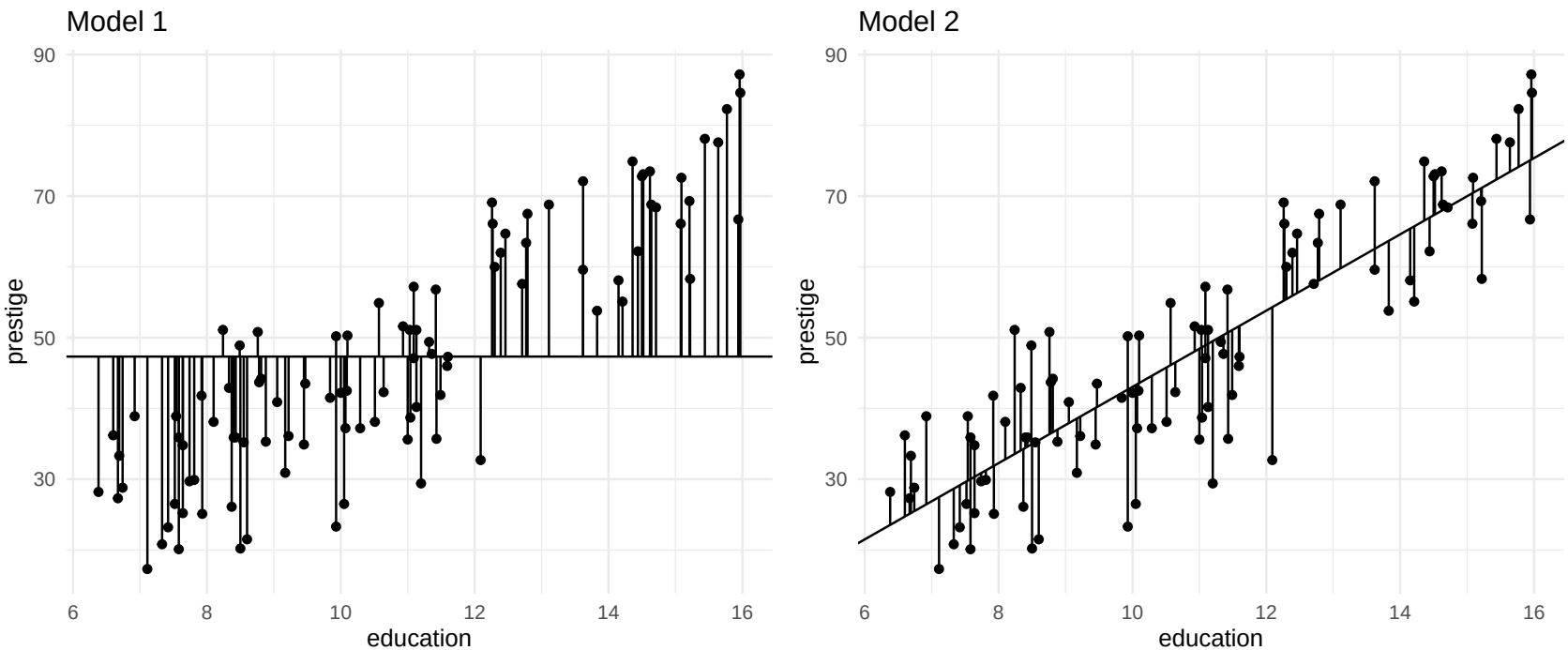
In this application we'll consider the following models:

- Model 1: `prestige ~ 1`
  - The intercept-only model $y_i = \beta_0 + \epsilon_i$
- Model 2: `prestige ~ education`
  - An alternative model $y_i = \beta_0 + \beta_1 \text{Educ}_i + \epsilon_i$
- Model 3: `prestige ~ education + type`
  - An even larger model
    $y_i = \beta_0 + \beta_1 \text{Educ}_i + \beta_2(\text{type} = \text{“prof"}) + \beta_3(\text{type} = \text{“wc"}) + \epsilon_i$

Questions: Is model 1 nested within model 3? Is model 2 nested within model 3? Why?

Example (cont.)

The F-test compares models using their *sums of squares* (squared residuals):



Recall: $SS = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ and $F = \dfrac{(SS_0 - SS_1)/(d_1 - d_0)}{SS_1/(n - d_1)}$

Example (cont.)

Below we use `anova()` to evaluate whether Model 2 is superior to Model 1:

```
## Analysis of Variance Table
##
## Model 1: prestige ~ 1
## Model 2: prestige ~ education
##   Res.Df     RSS Df Sum of Sq      F    Pr(>F)
## 1     97 28346.9
## 2     96  7064.4  1     21283 289.21 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
```

The p-value of the F-test is very small, thereby providing strong evidence that Model 2 better fits the data

Example (cont.)

We can repeat the same process to evaluate whether Model 3 is superior to Model 2:

```
## Analysis of Variance Table
##
## Model 1: prestige ~ education
## Model 2: prestige ~ education + type
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1     96 7064.4
## 2     94 5740.0  2    1324.4 10.844 5.787e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
```

Here we conclude that Model 3 is indeed superior to Model 2

Example (cont.)

An important property of the F-test is that it can simultaneously test many coefficients, including those corresponding to several different explanatory variables:

```
## Analysis of Variance Table
##
## Model 1: prestige ~ education
## Model 2: prestige ~ education + type + income + women
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1     96 7064.4
## 2     92 4679.0  4    2385.4 11.726 9.724e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
```

Here we see strong statistical evidence that the four-predictor model fits the data better than the model using only `education`.

Example (cont.)

What if we add a variable, `RX`, consisting of randomly generated values with no relation to the response?

```
## Analysis of Variance Table
##
## Model 1: prestige ~ education + type
## Model 2: prestige ~ education + type + RX
##   Res.Df  RSS Df Sum of Sq     F Pr(>F)
## 1     94 5740
## 2     93 5739  1    1.0515 0.017 0.8964
```

The F-test indicates a lack of evidence that this variable improves the model, so we should omit it as the complexity it adds to the model isn't warranted.

R^2 vs. Adjusted R^2

- In our previous example, the model using `RX` has an R^2 of 0.801, while the model without it has an R^2 of 0.797
  - Taking this further, if we add 6 predictors like `RX` that are generated to be completely unrelated to the outcome, the R^2 increases to 0.810
- This illustrates a problem with R^2, it *never* decreases when new predictors are added, so a larger R^2 doesn't necessarily mean a better model
  - *Adjusted* R^2 penalizes R^2 when more variables are included in the model (larger k)

$$R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n-1)}{n-k-1}$$

R^2 vs. Adjusted R^2

To summarize the F-test and R^2 vs. adjusted R^2 when adding a new predictor to a model:

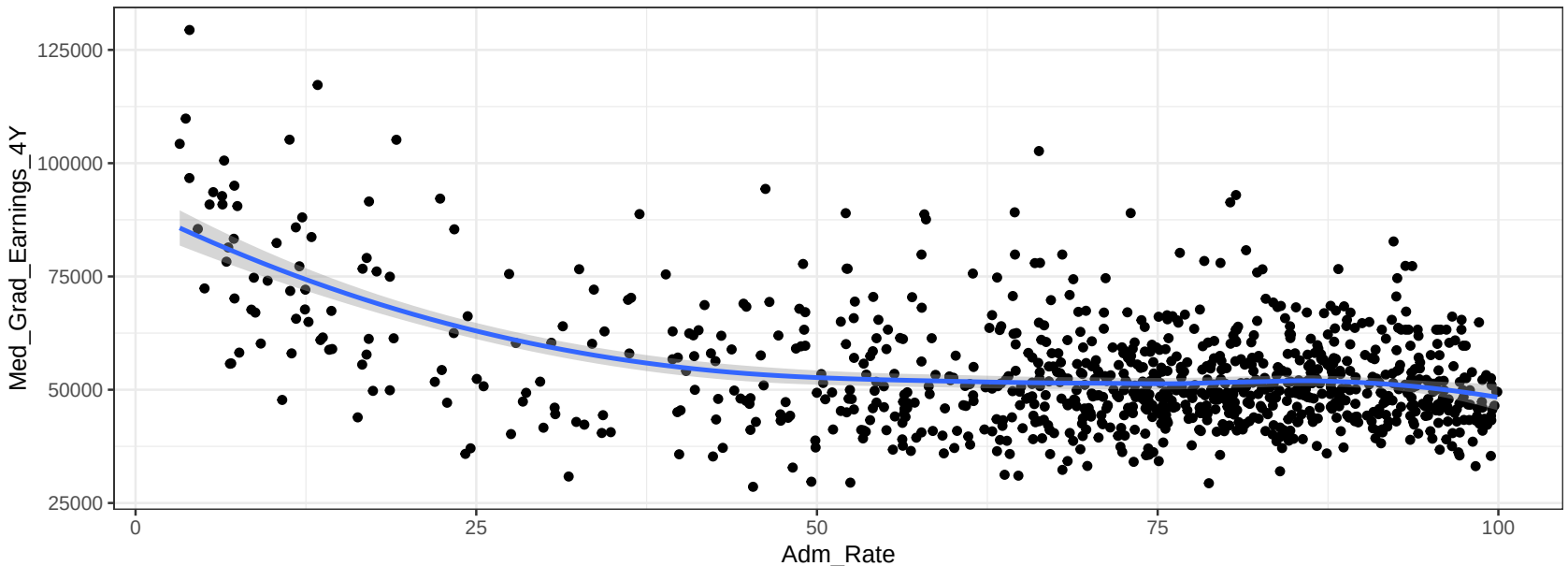
- New predictor is useful:
  - F-test yields a small p-value
  - Adjusted R^2 increases
- New predictor is not useful:
  - F-test yields a large p-value
  - Adjusted R^2 stays the same or decreases (even if R^2 increases)

Transformations

- The F-test assumes independent residuals with approximately constant variance and an approximately normal distribution
  - These assumptions are typically assessed using diagnostic plots from the larger model
- When the distribution of residuals appears skewed, log-transforming the outcome variable is a common strategy
  - This transformation can make the F-test assumptions more reasonable (and thus produce a more reliable p-value), but the model becomes more difficult to interpret

Log-transformation Example

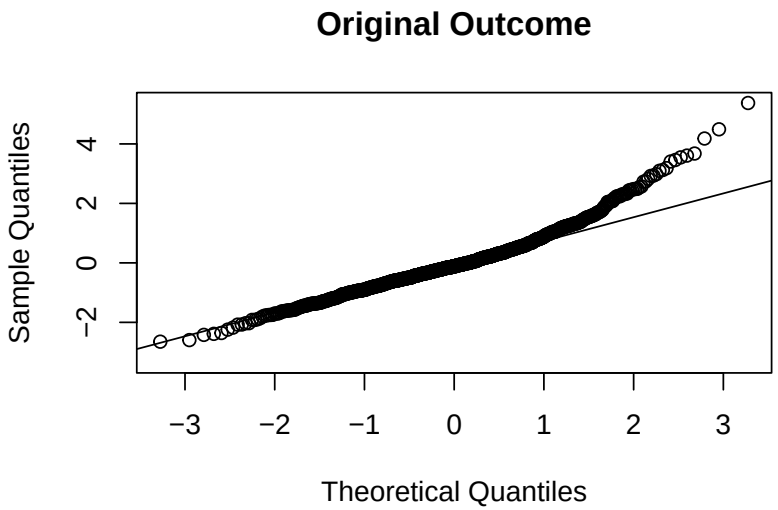
Consider the relationship between a college's admission rate and the median earnings of its graduates four years after graduation:

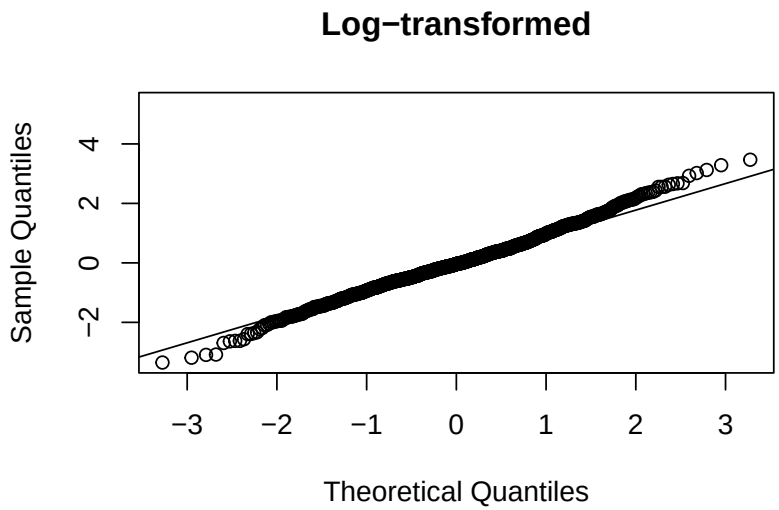


Most colleges have earnings between \$30k and \$60k, but there is a long right tail of colleges with median earnings over \$100k.

Log-transformation Example

The skew in the outcome is seen in the QQ-plot of residuals when the original outcome is used, but the skew lessens when the outcome is log-transformed





Log-transformation Example

For the original outcome, the fitted model equation is:

$$\widehat{\text{Earnings}} = 68210 - 211.1 \cdot \text{AdmRate}$$

For the log-transformed outcome, the fitted model equation is:

$$log(\widehat{\text{Earnings}}) = 11.1 - 0.003 \cdot \text{AdmRate}$$

If we undo the transformation:

$$\widehat{\text{Earnings}} = e^{11.1 - 0.003 \cdot \text{AdmRate}}$$

So, a 1-unit increase in admissions rate is expected to change median graduate earnings by a *multiplicative factor* of $e^{-0.003} = 0.997$

Conclusion

- The F-test allows us to statistically compare the fit of nested regression models. This has a variety of use cases, with the biggest being:
  - Testing the overall effect of adding a categorical predictor involving multiple categories
  - Testing whether a quadratic effect is needed
- We cannot use an F-test to determine if a log-transformation is warranted, as the transformation fundamentally changes the model fit by changing the units of the outcome
  - We assess the need for a log-transformation using residual diagnostic plots
- Additionally, adjusted R^2 is unit-less, so it provides a general way to compare model fits